

Practical 4: Binary Regression Modeling Practice

First author name goes here

You can have more than one author

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Data: Housing Loans

Research Question

Your goal is to plan and fit a regression model with `action_taken` as the response variable (using all data or Kent County only - you choose). You may choose predictors to match the analysis in an analysis of similar nationwide data by *The Markup* (see BIG HINT below), or if you prefer, make your own plan.

Data Details

This data is from HDMA, on mortgage approvals in Michigan in 2020. It is similar to, but not the same as, the data analyzed by The Markup.

The data were obtained from <https://ffiec.cfpb.gov/data-browser/> and pre-processed to make the file size smaller, and to filtering the data like The Markup did to include potential borrowers meeting conditions:

- Conventional mortgage only
- For purpose of home purchase
- 1- to 4-family home
- Site-built (not manufactured/mobile home)
- Not for business purpose
- “First lien” status
- Loan approved (“originated”) or denied

The code below reads in the data, and also does a bit of data wrangling that will be useful (you don’t have to understand details or be able to replicate this).

```
hdma_mi_20 <-  
  read_csv('https://sldr.netlify.app/data/hdma-mi-20.csv',  
           show_col_types = FALSE) |>  
  # race variable combining rare groups  
  mutate(  
    derived_race = case_when(  
      derived_race == 'White' ~ 'White',
```

```

    derived_race == 'Black or African American' ~ 'Black or African American',
    derived_race == 'Asian' ~ 'Asian',
    derived_race == 'Race Not Available' ~ 'Race Not Available',
    TRUE ~ 'Other'
  ),
  # make sure values of numeric values are numbers
  loan_to_value_ratio = parse_number(loan_to_value_ratio),
  property_value = parse_number(property_value),
  loan_term = parse_number(loan_term),
  # whether there is a co-applicant
  co_applicant = ifelse(`co-applicant_ethnicity-1` == 5, 'No', 'Yes'),
  # service underwriting the mortgage
  aus = case_when(`aus-1` == 1 ~ 'Desktop Underwriter (DU)',
                  `aus-1` == 2 ~ 'Loan Prospector (LP) or Loan Product Advisor',
                  `aus-1` == 1111 ~ 'Exempt',
                  TRUE ~ 'Other'),
  # unknown age is encoded as "8888"
  applicant_age = ifelse(applicant_age == '8888', NA, applicant_age)
)

```

```

Warning: There were 3 warnings in `mutate()`.
The first warning was:
i In argument: `loan_to_value_ratio = parse_number(loan_to_value_ratio)`.
Caused by warning:
! 3489 parsing failures.
  row col expected actual
    77  -- a number Exempt
   9067  -- a number Exempt
   9068  -- a number Exempt
   9069  -- a number Exempt
   9070  -- a number Exempt
  ....
See problems(...) for more details.
i Run `dplyr::last_dplyr_warnings()` to see the 2 remaining warnings.

```

the warnings about “parsing failures” are ok: those are cases where numeric variables had non-numeric contents which R failed to convert to numbers (this was expected).

```

# Limit to only Kent County (where Calvin is located), if desired
hdma_kent_20 <- hdma_mi_20 |>
  filter(county_code == 26081) |>
  # at the county scale, it is very rare
  # for race to be known and sex to be unknown
  # and this causes trouble...so remove cases where sex is not known
  filter(derived_sex != 'Sex Not Available')

```

Minimal Data Prep

Note that in this housing dataset, many categorical variables are stored as numeric codes.

Even if you don't want to do *any* additional data prep, make sure you `factor()` your response variable, plus any `factor()` any other *categorical* predictors you want to use in your model, so they are not treated as numeric data.

Something like:

```
hdma_mi_20 <- hdma_mi_20 |>
  mutate(my_response_variable = factor(my_response_variable),
         my_predictor_1 = factor(my_predictor_1),
         my_predictor_2 = factor(my_predictor_2)
         # (repeat above for as many categorical predictors as you need to use)
  )
```

(Optional) More Advanced Data Prep

If you want to do more...ask a helper!

You can find a list of variable definitions and codes used in the variables in the dataset at:

<https://ffiec.cfpb.gov/documentation/publications/loan-level-datasets/lar-data-fields> (search and find is your friend!)

Ideally you will create and use variables where the category labels are intelligible words rather than numeric codes! You may also need to rename variables that have “-” in the names - refer to the “old/bad” names in back-ticks like: `bad-variable-name`. For detailed examples on how to rename and reorganize this dataset for use, see: <https://stacyderuiter.github.io/stat245-sp25/housing-wrangling.html>.

Tasks

1. **Plan:** Articulate your model plan. You may do this one of two ways:
 - A. Make your own plan, considering dataset size and maybe a causal diagram; **or**,
 - B. To replicate (as closely as possible) the model fitted in the analysis by The Markup, relying on their expertise in this field. *To do this: See the “BIG HINT” below.*
2. **(Optional) Explore** Read your data into R and make a one well-designed exploratory plot of the data that is relevant to your research question. Make note of any patterns you see in the graph.
3. **Fit** a regression model that is consistent with your model plan. Show its `summary()`.
 - Note: depending on the dataset and predictors you use, you may well run into model-fitting warnings. *Don't ignore these warnings*; work with your team or instructors to solve them!
4. **Assess** your model, showing all your work: clearly state which condition you're checking, whether it's met, and what evidence supports your decision.

5. State your **conclusions** about the relationship between your key predictor of interest and the response. To support your answer, you might refer to **model comparison/selection** results, **model assessment** (to confirm results are trustworthy) and maybe a **prediction plot** or data graph to describe effect size/direction.

BIG HINT: The Markup's analysis

In the article we recently read, a box was included linking to a detailed description of the methods Markup analysts used to obtain their results. They state that “Our national model for conventional applications contained 17 variables:

- Race (our variables `derived_race` and/or `derived_ethnicity`)
- Sex (our variable `derived_sex`)
- Whether the application had a co-applicant (`co_applicant`)
- Age (`applicant_age`)
- Income (`income`)
- Loan amount (`loan_amount`)
- Property value (`property_value`)
- Mortgage term (`loan_term`, in months)
- Credit model used (`applicant_credit_score_type`)
- Debt-to-income ratio (`debt_to_income_ratio`)
- Combined loan-to-value ratio (`loan_to_value_ratio`)
- The automated underwriting system used (`aus`)
- The ratio between the median income of the census tract where the property is located and the median income of the metro area (***not available to us***)
- The type of lender (added variable to HMDA data) (***not available to us***)
- The size of the lender (added variable to HMDA data) (***not available to us***)
- Non-Hispanic White population percentage of the census tract where the property is located (added variable to HMDA data) (***not available to us***)
- Size of the metro area where the property is located (added variable to HMDA data) (***not available to us***)